Random Projection for Indirect Quantization

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***Abstract*—In indirect quantization tasks, the quantize-then-estimate (QE) scheme is commonly adopted when encoders are distributed or task-agnostic. In some scenarios, however, the observation dimension substantially exceeds that of the underlying state, rendering direct quantization of the observations highly inefficient. In this work, we propose a random-projection-then-quantization (RPQ) scheme. Specifically, high-dimensional observations are first projected onto a lower-dimensional representation via random projection, after which quantization is applied only to the reduced dimensions rather than to all original dimensions. This strategy reduces the required bit rate and thereby alleviates communication overhead. The rate–distortion performance of the proposed scheme is analyzed. Under low-rate constraints, the proposed scheme outperforms the conventional QE scheme and, in certain scenarios, continues to do so even at higher rates.**

***Index Terms*—Composite source, indirect quantization problem, random projection, rate-distortion theory.**

I. Introduction

The problem of semantic communication has long attracted attention [1], [2]. Recent studies have incorporated semantic information into the design of communication systems [3], [4]. These works are typically task-oriented, rather than aimed at directly compressing or transmitting signals, with objectives closely related to semantic information. A common model for studying semantic communication is the composite source model [5]–[7], which consists of an intrinsic state and an extrinsic observation. The intrinsic state represents the semantic information, whereas the encoder has access only to the observation. Since many practical problems involve estimating continuous latent parameters (e.g., channel coefficients, localization coordinates, phase offsets, or other physical quantities), modeling the semantic state S as a continuous random vector is both natural and necessary in such settings. In this work, the composite source model with a continuous-valued state is adopted for further investigation.

Our problem falls into the category of indirect source coding [8]–[10]. In [7], [8], the optimality of the estimate-then-compress (EC) scheme was established. In the context of quantization, the corresponding approach is the estimate-then-quantize (EQ) scheme [11], in which the state is first estimated from the observation and then quantized.

Nevertheless, in many practical applications, encoders need to be universal and task-agnostic, i.e., independent of both the observation distribution and the underlying semantic relationship, making the EQ scheme infeasible. A natural alternative is the quantize-then-estimate (QE) scheme. In [12], [13], the rate–distortion risk of the compress-then-estimate (CE) scheme was analyzed, and a method for designing estimators independent of specific compression techniques was proposed.

In some cases, although the exact relationship between the state and the observation is unknown, the observation is known to have a much higher dimension than the state. In such highly overdetermined systems, directly quantizing the observation wastes rate. However, dimensionality reduction techniques such as principal components analysis (PCA) are distribution-dependent and require prior knowledge of the source statistics, violating the constraint of being universal. To address this, we propose a random-projection–then–quantization (RPQ) scheme: the high-dimensional observation is first projected into a lower-dimensional representation and then quantized. A task-specific estimator then reconstructs the state from the quantized low-dimensional representation.

As related work, random projection has been used in signal processing for practical tasks such as computer vision [14], [15] and speech recognition [16]. These works empirically demonstrate its potential, e.g., reducing transmitted data or improving computational efficiency while maintaining classification accuracy, but do not investigate it from a rate–distortion perspective. Random projection has also been used in compressed sensing [17], which aims to recover high-dimensional sparse signals. Some works in the machine learning context use randomly projected and quantized vectors to estimate inner products [18] or provide labels for supervised learning [19]. None of these works investigate quantization with random projections in the context of indirect source coding from a rate–distortion perspective. In this work, we explicitly adopt the composite source model to formulate and analyze this problem. Different from the prior works, our formulation imposes no sparsity assumptions on the signals and treats the encoder as task-agnostic and distribution-agnostic.

Our main contributions are as follows:

- We propose the RPQ scheme and derive its high-rate rate-distortion performance for composite sources under mild regularity conditions, given a fixed projection matrix.
- We establish the relationship between the expected estimation distortion and the dimension K of the low-dimensional representation vector for a general linear Gaussian composite source with a scalar state and isotropic noise, and analyze the trade-off between estimation distortion and distortion induced by quantization.

- Through numerical results, we compare the rate–distortion performance of the RPQ scheme for different values of K with those of the QE and EQ. The results show that random projection improves rate–distortion performance at low rates and, in some scenarios, even at higher rates.

The remaining part of the paper is organized as follows: In Section II, we describe the source model adopted in this work and formulate the problem under consideration. Section III presents the RPQ scheme and analyzes its performance, including the rate–distortion performance in the high-rate regime for a given projection matrix Φ. In Section IV, we establish the relationship between the expected estimation distortion and the dimension K of the low-dimensional representation in a commonly studied setting, and analyze the trade-off between estimation distortion and distortion induced by quantization. Section V compares the rate–distortion performance of the RPQ scheme under different values of K with those of the EQ and QE schemes via numerical results. Finally, Section VI concludes the paper.

II. Problem Formulation

A composite source consists of an intrinsic state S and an extrinsic observation X, where each state value corresponds to a distinct sub-source and distribution of X. In encoding, the encoder observes only X [7], [20], and the state can only be inferred from it. In this paper, we focus exclusively on indirect quantization tasks, requiring only the reconstruction of S.

The optimality of the EC scheme under the mean-squared error (MSE) criterion was proved in [7], [8]. In the EC scheme, the state S is first estimated as $\tilde{S}$ from the observation X, which is then compressed into $\hat{S}$ and delivered to the destination. The procedure is illustrated in Figure 1.

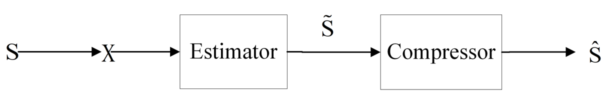


Fig. 1. The procedure of the EC scheme

When the encoder is task-agnostic, the EC scheme cannot be implemented. In this case, the CE scheme can be applied. Here, X is first compressed into $\hat{X}$, and at the destination, a task-specific estimator then reproduces $\hat{S}$ from $\hat{X}$. The procedure of this scheme is illustrated in Figure 2.

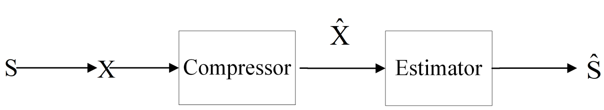


Fig. 2. The procedure of the CE scheme

In this paper, quantization is adopted as the compression method, and the MSE distortion is used as the distortion metric. Denoting the state distortion as d_S, we have $d_S(s,\hat{s}) = \|s-\hat{s}\|_2^2$. When designing the quantizer, our objective is to minimize $\mathbb{E}\{d_S(S,\hat{S})\}$. Although S cannot be observed by the quantizer, it turns out that, this indirect optimization objective can be equivalently expressed in a direct form [20] as $\hat{d}_S(x,\hat{s}) = \mathbb{E}\{d_S(S,\hat{s}) \mid X = x\}$.

Denote the quantizer and its input and output by $Q(\cdot)$, $\mathbf{U}$ and $\mathbf{V} = (V_1, V_2, \ldots)$, respectively. In some designs, the quantizer produces $\hat{X}$; in others, it outputs $\hat{S}$ directly. The rate is $R = H(\mathbf{V})$ under joint coding, and $R = \sum_{i=1}^{K} H(V_i)$ under independent coding (relying solely on marginal entropies), where K denotes the dimension of V.

III. The Random-Projection-then-Quantization Scheme

In indirect quantization, the conventional QE scheme becomes inefficient when the observation dimension n significantly exceeds the state dimension. To address this, we propose the RPQ scheme, which projects X into a K-dimensional ($K \le n$) representation $Y = \Phi X$ using a random matrix $\Phi \in \mathbb{R}^{K\times n}$ with i.i.d. Gaussian entries, shared by both the encoder and the decoder, thereby avoiding the limitations of both distribution-dependent methods and naive subsampling. Unlike PCA, which requires prior knowledge of the observation covariance, it is distribution-agnostic and task-agnostic. Moreover, rather than discarding dimensions, which may incur large performance variability and substantial loss of information about S, the projection combines all input dimensions linearly. The RPQ procedure, shown in Fig. 3, induces the Markov chain $S \leftrightarrow X \leftrightarrow Y \leftrightarrow \hat{Y} \leftrightarrow \hat{S}$.

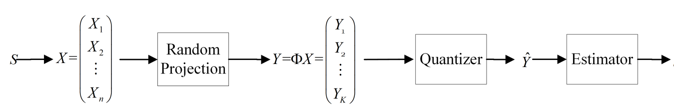


Fig. 3. The procedure of the RPQ scheme

First, consider the estimator design at the destination for a fixed Φ. When the estimator receives $\hat{Y} = \hat{y}_i$, where $\hat{y}_i$ corresponds to the i-th quantization cell $\mathcal{C}_i$, the optimal estimate $\hat{s}^*$ satisfies:

$$\hat{s}^* = \arg\min_{\hat{s}} \mathbb{E}\{\|\hat{s} - S\|_2^2 | Y \in \mathcal{C}_i\}. \tag{1}$$

Define $g(y) = \mathbb{E}\{S \mid Y = y\}$. We have

$$\begin{aligned} &\mathbb{E}\{\|\hat{s} - S\|_2^2 \mid Y \in \mathcal{C}_i\} \\ &= \mathbb{E}\{\|\hat{s} - g(Y) + g(Y) - S\|_2^2 \mid Y \in \mathcal{C}_i\} \\ &= \mathbb{E}\{\|\hat{s} - g(Y)\|_2^2 \mid Y \in \mathcal{C}_i\} + \mathbb{E}\{\|g(Y) - S\|_2^2 \mid Y \in \mathcal{C}_i\} \\ &+ 2\mathbb{E}\{(g(Y) - S)^T(\hat{s} - g(Y)) \mid Y \in \mathcal{C}_i\}. \end{aligned} \tag{2}$$

For the cross term in (2),

$$\begin{aligned} &\mathbb{E}\{(g(Y) - S)^T(\hat{s} - g(Y)) \mid Y \in \mathcal{C}_i\} \\ &= \mathbb{E}_Y\{\mathbb{E}_S\{(g(Y) - S)^T(\hat{s} - g(Y)) \mid Y\} \mid Y \in \mathcal{C}_i\} \\ &= \mathbb{E}_Y\{\mathbb{E}_S\{g(Y) - S \mid Y\}^T(\hat{s} - g(Y)) \mid Y \in \mathcal{C}_i\}. \end{aligned} \tag{3}$$

Since $g(Y) = \mathbb{E}_S\{S \mid Y\}$, the cross term in (2) vanishes.

Let $L_i(\hat{s}) = \mathbb{E}\{\|\hat{s} - g(Y)\|_2^2 | Y \in \mathcal{C}_i\}$. Then (1) is equivalent to $\hat{s}^* = \arg\min_{\hat{s}} L_i(\hat{s})$, where $L_i(\hat{s}) = \hat{s}^T\hat{s} - 2\mathbb{E}\{g(Y)|Y \in \mathcal{C}_i\}^T\hat{s} + \mathbb{E}\{g(Y)^T g(Y)|Y \in \mathcal{C}_i\}$. Setting $\frac{\partial L_i}{\partial \hat{s}} = 0$ gives

$$\hat{s}^* = \mathbb{E}\{g(Y)|Y \in \mathcal{C}_i\}. \tag{4}$$

Consider continuously differentiable g, and assume the quantizer satisfies $\hat{y}_i = \mathbb{E}\{Y|Y \in \mathcal{C}_i\}$ for each cell $\mathcal{C}_i$. If $g(y)$ is linear, then $\mathbb{E}\{g(Y)|Y \in \mathcal{C}_i\} = g(\hat{y}_i)$, and $\hat{S} = g(\hat{Y})$ is optimal. For non-linear g, in the high-rate regime, when Y lies in a shrinking quantization cell $\mathcal{C}_i$, we have

$$g(Y) = g(\hat{y}_i) + J_g(\hat{y}_i)(Y - \hat{y}_i) + o(\|Y - \hat{y}_i\|_2), \tag{5}$$

where $J_g(\hat{y}_i)$ denotes the Jacobian matrix of g at $\hat{y}_i$. The shrinking quantization cell ensures that the remainder term vanishes for $Y \in \mathcal{C}_i$. Hence,

$$\begin{aligned}\mathbb{E}\{g(Y)|Y \in \mathcal{C}_i\} &\approx \mathbb{E}\{g(\hat{y}_i) + J_g(\hat{y}_i)(Y - \hat{y}_i)|Y \in \mathcal{C}_i\} \\ &= g(\hat{y}_i).\end{aligned} \tag{6}$$

Therefore, the estimator $\hat{S} = g(\hat{Y})$ is asymptotically optimal in the high-rate regime.

In this paper, we adopt the estimator $\hat{S} = g(\hat{Y})$, even when neither the linearity nor the high-rate condition holds, or when the quantizer does not strictly satisfy the centroid condition $\hat{y}_i = \mathbb{E}\{Y|Y \in \mathcal{C}_i\}$.

For a given Φ,

$$\mathbb{E}\{\|S - \hat{S}\|_2^2\} = D_0(\Phi) + \mathbb{E}\{\|g(Y) - g(\hat{Y})\|_2^2\}, \tag{7}$$

where $D_0(\Phi) = \mathbb{E}\{\|S - g(Y)\|_2^2\}$ is the estimation distortion, and $\mathbb{E}\{\|g(Y) - g(\hat{Y})\|_2^2\}$ is the distortion induced by quantization.

Theorem 1. *For a composite source and a random projection matrix Φ, if $g(y)$ is twice continuously differentiable and all first and second-order derivatives bounded, then in the high-rate regime, where the quantization cells are sufficiently small such that the PDF $f_Y(y)$ is approximately uniform within each cell, the distortion-rate performance of the RPQ scheme with a uniform scalar quantizer and estimator $\hat{S} = g(\hat{Y})$ is approximated by*

$$D \approx D_0(\Phi) + \frac{\mathbb{E}\{\mathrm{tr}[W(Y)]\}}{12} 2^{\frac{2}{K}h(Y)} 2^{-\frac{2}{K}R}, \tag{8}$$

where $h(\cdot)$ is the differential entropy and $W(y) = J_g(y)^T J_g(y)$. For independent marginal entropy coding, $h(Y)$ in (8) is replaced by $\sum_{i=1}^K h(Y_i)$.

Proof. See Appendix A. □

Consider a composite source where $S \in \mathbb{R}^m$ and $X \in \mathbb{R}^n$ are zero-mean and jointly Gaussian, related by $X = HS + W$, with H an $n \times m$ matrix and $W \sim \mathcal{N}(\mathbf{0}, \Sigma_W)$ independent of S. Denote the joint covariance matrix of (S, X) as

$$\begin{bmatrix} \Sigma_S & \Sigma_{SX} \\ \Sigma_{SX}^T & \Sigma_X \end{bmatrix}. \tag{9}$$

We have $\Sigma_{SX} = \Sigma_S H^T$ and $\Sigma_X = H\Sigma_S H^T + \Sigma_W$.

For a fixed Φ, S and Y are jointly Gaussian and the relation can be written as $S = BY + N$, where $B = \Sigma_{SY}\Sigma_Y^{-1}$ and N is independent of Y with distribution $\mathcal{N}(\mathbf{0}, \Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T)$. Here Σ_{SY} and Σ_Y denote the cross-covariance matrix between S and Y, and the covariance matrix of Y, respectively.

Corollary 1. *For the jointly Gaussian composite source described above, the linear estimator $\hat{S} = g(\hat{Y}) = B\hat{Y}$ is optimal. In this case, the weighting matrix $W(y)$ is constant for all y, i.e., $W(y) = B^T B$. The high-rate distortion–rate function in Theorem 1 then simplifies to*

$$D \approx D_0(\Phi) + \frac{\|B\|_F^2}{12} 2^{\frac{2}{K}h(Y)} 2^{-\frac{2}{K}R}, \tag{10}$$

with $\|\cdot\|_F$ denoting the Frobenius norm. Similarly, under independent marginal entropy coding, $h(Y)$ is substituted by $\sum_{i=1}^K h(Y_i)$.

IV. Case Study

Having established a high-rate characterization under mild regularity conditions in the previous section, we now consider a specific source model to gain analytical insight through closed-form results. For a given K, we analyze the performance by taking the expectation over the random projection matrix Φ, whose entries are assumed to be i.i.d. $\mathcal{N}(0, \frac{1}{K})$. We then study the impact of K on the rate–distortion performance.

Consider a composite source with a scalar state $S \sim \mathcal{N}(0, 1)$ and an n-dimensional observation $X = hS + W$, where h is an n-dimensional vector and $W \sim \mathcal{N}(\mathbf{0}, \sigma_W^2 I_n)$ is independent of S.

Using (7), taking the expectation over Φ yields $\mathbb{E}_\Phi\{D\} = \mathbb{E}_\Phi\{D_0(\Phi)\} + \mathbb{E}_\Phi\{\mathbb{E}\{\|BY - B\hat{Y}\|_2^2\}\}$.

Theorem 2. *For the composite source above, projecting X into a K-dimensional vector Y using a Gaussian random matrix Φ yields the expected estimation distortion*

$$\mathbb{E}_\Phi\{D_0(\Phi)\} = \mathbb{E}\left\{\frac{\sigma_W^2}{\sigma_W^2 + \|h\|_2^2 l}\right\}, l \sim \mathrm{Beta}\left(\frac{K}{2}, \frac{n-K}{2}\right),$$

where $\mathrm{Beta}(\cdot,\cdot)$ denotes the Beta distribution.

Proof. See Appendix B. □

Remark 1. *Theorem 2 indicates that increasing K shifts the density of l toward one, making l more concentrated near 1 and reducing the expected estimation distortion.*

A special case arises when $K = n$:

$$\begin{aligned}\mathbb{E}\{D_0(\Phi)\} &= \mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T] \\ &= \mathrm{tr}[\Sigma_S - \Sigma_{SX}\Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi\Sigma_{SX}^T].\end{aligned}$$

Since Φ is invertible with probability one [21],

$$\mathbb{E}\{D_0(\Phi)\} = \mathrm{tr}[\Sigma_S - \Sigma_{SX}\Sigma_X^{-1}\Sigma_{SX}^T] \quad \text{a.s.} \tag{11}$$

Thus, for $K = n$, the expected estimation distortion equals that of directly estimating S from X.

Theorem 1 gives the high-rate distortion–rate performance of the RPQ scheme. When R exceeds a threshold, (8) is dominated by $D_0(\Phi)$. Intuitively, since Φ is full rank with probability one, a larger K preserves more information about X, reducing $D_0(\Phi)$. Theorem 2 formally establishes this for the composite source considered. Hence, in this regime, dimensionality reduction cannot reduce the total distortion.

Nevertheless, when R does not exceed the threshold, the second term in (7) cannot be neglected. Increasing K lowers the rate per dimension, enlarging scalar quantization intervals and thus increasing quantization distortion. Hence, varying K introduces a trade-off between estimation and quantization distortions. We emphasize that the Gaussian model is used only for closed-form expressions. The trade-off itself does not depend on Gaussianity and is not specific to this model.

V. Numerical Results

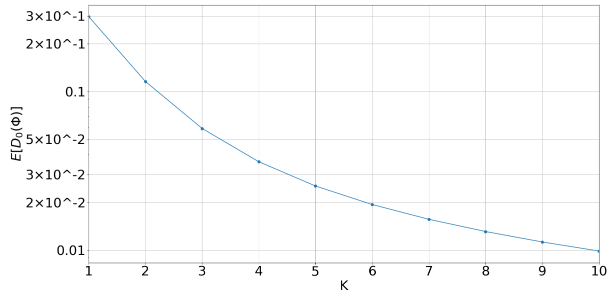
In this section, we evaluate the RPQ scheme for different K and compare it with the QE and EQ schemes. We first conduct numerical simulations on four instances of the source models from Section IV. For the first two sources, we set $n = 10$ and $\sigma_W^2 = 0.1$ and 0.01. For the other two sources, $n = 50$ with $\sigma_W^2 = 0.1$ and 0.01. For all of the four sources, $h = \mathbf{1}_n$, where $\mathbf{1}_n$ denotes the n-dimensional all-one vector. As in Section IV, the entries of Φ are i.i.d. and follow $\mathcal{N}(0, \frac{1}{K})$.

Figure 4 shows the relation between the expected estimation distortion $\mathbb{E}_\Phi\{D_0(\Phi)\}$ and the retained dimension K for the four source models. The distortion decreases as K increases. When the observation dimension is large, or the noise is small (i.e., high observation redundancy), it drops rapidly. As K becomes large, however, retaining more dimensions provides a very marginal additional reduction in estimation distortion. Thus, in many scenarios, random dimensionality reduction can reduce the bit rate while still meeting distortion constraints, unless extremely high precision is required.

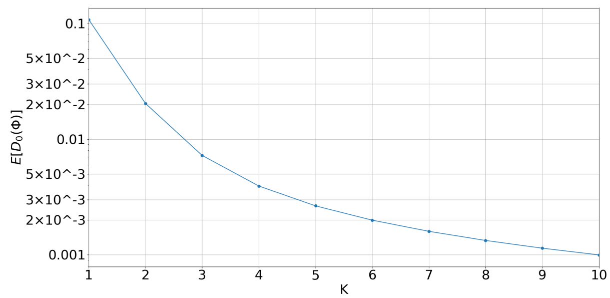
When the rate is not sufficiently high, the second term in (7) cannot be neglected. Increasing K reduces the bit rate allocated to each dimension, thereby expanding the quantization intervals of the scalar quantizer and consequently increasing quantization distortion. Thus, varying K introduces a trade-off between estimation distortion and quantization distortion.

Figure 5 compares the performance of the RPQ scheme for different K with the QE and EQ schemes, where uniform scalar quantization is used in all schemes. For each K, 10,000 instances of Φ are randomly generated, and for each Φ, 50,000 pairs of (S, X) are simulated. At the quantizer, the number of bits allocated to each dimension is the same. The total rate is calculated as the sum of marginal entropies of the quantization results.

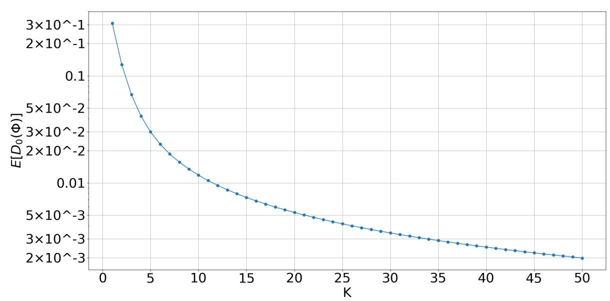
The results show that the EQ scheme outperforms the other schemes, indicating that the knowledge of the semantic relationship between the observation and the state improves the quantizer's rate–distortion performance. However, when the quantizer needs to be task-agnostic, the EQ scheme cannot be implemented. In the low-rate regime, the distortion induced by quantization becomes more significant. Thus, although the estimation distortion increases, employing random projection



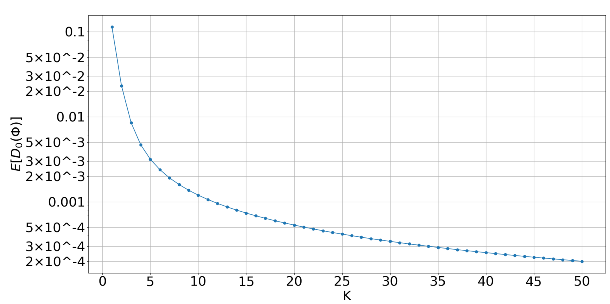
(a): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 10, \sigma_W^2 = 0.1$.



(b): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 10, \sigma_W^2 = 0.01$.

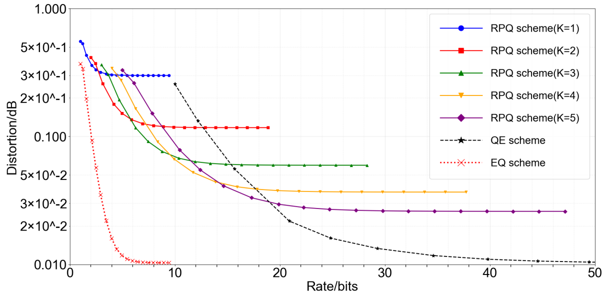


(c): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 50, \sigma_W^2 = 0.1$.

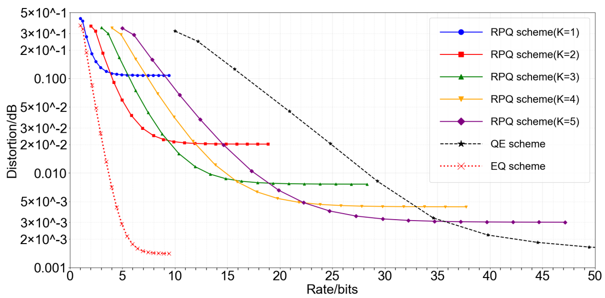


(d): $\mathbb{E}\{D_0(\Phi)\}$ vs K when $n = 50, \sigma_W^2 = 0.01$.

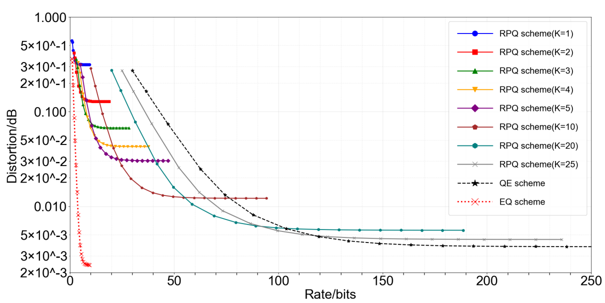
Fig. 4. Relationship between expected estimation distortion $\mathbb{E}\{D_0(\Phi)\}$ and the retained dimension K.



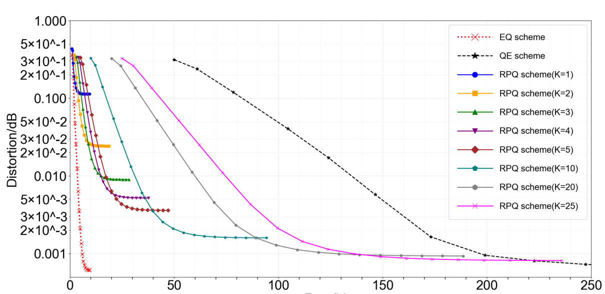
(a): The distortion-rate performances of different schemes when $n = 10, \sigma_W^2 = 0.1$.



(b): The distortion-rate performances of different schemes when $n = 10, \sigma_W^2 = 0.01$.



(c): The distortion-rate performances of different schemes when $n = 50, \sigma_W^2 = 0.1$.



(d): The distortion-rate performances of different schemes when $n = 50, \sigma_W^2 = 0.01$.

Fig. 5. The distortion-rate performances of different schemes.

substantially reduces the total distortion due to the higher bit rate allocated to each scalar quantizer dimension.

Notably, when n is large or the noise is small, the RPQ scheme can still outperform the QE scheme even at higher rates. Intuitively, the information about the intrinsic state is mostly concentrated within a low-dimensional subspace of the high-dimensional observation space. Retaining all dimensions results in a lower bit rate allocated to each dimension, while the dimensions often exhibit significant redundancy. By projecting observations into a low-dimensional representation, the bits are allocated among fewer dimensions with less redundancy. As shown in the numerical results, in this regime, quantization still has a significant impact on the total distortion.

Another trend observed is that, for the same source, the optimal K increases with the rate, which highlights the trade-off between the two types of distortions.

The second simulation is conducted to further test whether a single, task-agnostic random matrix can improve rate-distortion performance across different inference objectives. Consider a composite source with a 2-dimensional state $S = (S_1, S_2)$, where

$$S \sim \mathcal{N}(\mathbf{0}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}), \tag{12}$$

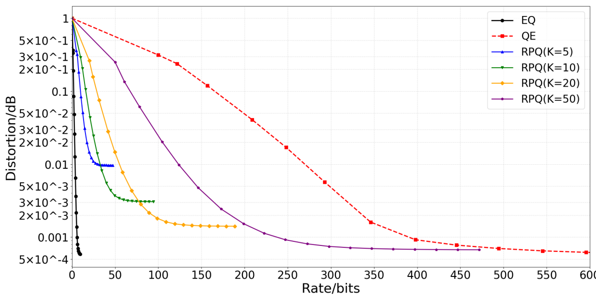
and an observation $X = h_1 S_1 + h_2 S_2 + W$, where h_1 and h_2 are n-dimensional vectors and $W \sim \mathcal{N}(\mathbf{0}, \Sigma_W)$ independent of S. This gives rise to two inference tasks based on the same observation X. Specifically, the tasks are to reconstruct S_1 and S_2 from the observation, respectively.

In our simulation, $n = 100$, $\rho = 0$, $\Sigma_W = 0.01 I_n$, $h_1 = (1, 0, 1, 0, \dots)$ and $h_2 = (0, 1, 0, 1, \dots)$. This setting is chosen to represent scenarios where task-relevant information is distributed across different signal components. For each Monte-Carlo trial, a task-agnostic RPQ encoder utilizes a random projection matrix $\Phi \in \mathbb{R}^{K \times n}$ with i.i.d. entries $\mathcal{N}(0, 1/K)$. Separate estimators then reconstruct S_1 and S_2 from the same quantized measurement $\hat{Y}$, evaluating the rate-distortion efficiency of RPQ for multiple downstream tasks.

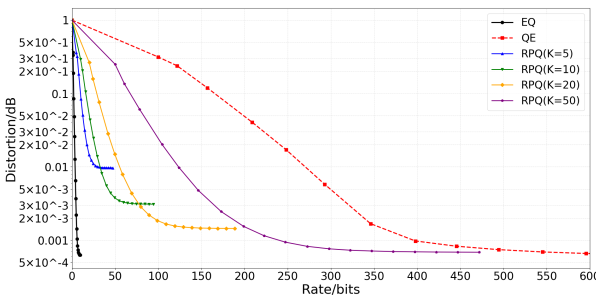
Figure 6 compares the performance of the RPQ scheme for different K with the QE and EQ schemes in reconstructing S_1 and S_2, respectively. The results show that RPQ improves the rate–distortion performance for both tasks simultaneously, despite being strictly task-agnostic. Unlike simple subsampling, which can be highly sensitive to the selected components, RPQ linearly mixes all signal components before quantization. In this setting, for example, equidistant subsampling with a stride of 2 ($K = 50$) can completely miss the informative components of one task, whereas RPQ is less vulnerable to such component mismatch.

VI. Conclusion

We propose the RPQ scheme to mitigate the rate inefficiency of conventional QE caused by system overdetermination in certain indirect quantization problems. Its rate-distortion analysis reveals a fundamental trade-off between



(a): the distortion–rate performance for reconstructing S_1.



(b): the distortion–rate performance for reconstructing S_2.

Fig. 6. The distortion-rate performances of different schemes for the two tasks.

estimation distortion and quantization distortion, modulated by the retained dimension K. The expected estimation distortion is characterized analytically as a function of K. Numerical results show that RPQ outperforms QE in the low-rate regime, and also at higher rates when the observation dimension is large or the noise is small. In future work, we plan to study the relationship between estimation distortion and the dimension K in more general source models. Another promising direction is to investigate how, if a feedback channel is available, both the retained dimension K and the projection matrix can be adaptively adjusted when the encoder is unaware of the exact relationship between the state and the observation.

APPENDIX A

For any estimator of the form $g(y) = \mathbb{E}\{S|Y = y\}$, the total distortion can be decomposed as:

$$\mathbb{E}\{\|S-\hat{S}\|_2^2\} = \mathbb{E}\{\|S-g(Y)\|_2^2\}+\mathbb{E}\{\|g(Y)-g(\hat{Y})\|_2^2\}. \tag{13}$$

Under high-rate condition, through (5), we have

$$\mathbb{E}\{\|g(Y) - g(\hat{Y})\|_2^2\} \approx \mathbb{E}\{\|J_g(\hat{Y})(Y - \hat{Y})\|_2^2\}. \tag{14}$$

Consequently, the part of distortion induced by quantization in the indirect problem is transformed into a direct distortion measured under a weighted squared error (WSE) criterion, characterized by the weight matrix $W(y) = J_g(y)^T J_g(y)$. So

$$\mathbb{E}\{\|S-\hat{S}\|_2^2\} = D_0(\Phi)+\mathbb{E}\{(Y-\hat{Y})^T W(\hat{Y})(Y-\hat{Y})\}. \tag{15}$$

Denote the second term as D_{quant}. We have

$$D_{quant} = \mathbb{E}_{\hat{Y}}\{\mathbb{E}_Y\{(Y - \hat{Y})^T W(\hat{Y})(Y - \hat{Y}) \mid \hat{Y}\}\}. \tag{16}$$

For the inner expectation

$$\begin{aligned} &\mathbb{E}_Y\{(Y - \hat{Y})^T W(\hat{Y})(Y - \hat{Y}) \mid \hat{Y}\} \\ &= \mathbb{E}_Y\{\mathrm{tr}[(Y - \hat{Y})^T W(\hat{Y})(Y - \hat{Y})] \mid \hat{Y}\} \\ &= \mathrm{tr}[\mathbb{E}_Y\{W(\hat{Y})(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y}\}] \\ &= \mathrm{tr}[W(\hat{Y})\mathbb{E}_Y\{(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y}\}]. \end{aligned} \tag{17}$$

For a given $\hat{Y} = \hat{y}_i$ in cell $\mathcal{C}_i$,

$$\begin{aligned} &\mathbb{E}_Y\{(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y} = \hat{y}_i\} \\ &= \int_{\mathcal{C}_i} (y - \hat{y}_i)(y - \hat{y}_i)^T \frac{f_Y(y)}{P(Y \in \mathcal{C}_i)} dy. \end{aligned} \tag{18}$$

Under high-resolution condition, $f_Y(y) \approx f_Y(\hat{y}_i), \forall y \in \mathcal{C}_i$. Hence $P(Y \in \mathcal{C}_i) \approx \mathrm{Vol}(\mathcal{C}_i) f_Y(\hat{y}_i)$, where $\mathrm{Vol}(\cdot)$ denotes the volume of the cell and $\mathrm{Vol}(\mathcal{C}_i)$ equals to Δ^K. So

$$\begin{aligned} &\mathbb{E}_Y\{(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y} = \hat{y}_i\} \\ &\approx \frac{1}{\Delta^K} \int_{\mathcal{C}_i} (y - \hat{y}_i)(y - \hat{y}_i)^T dy. \end{aligned} \tag{19}$$

Denote the j-th component of y and $\hat{y}_i$ as y_j and $\hat{y}_{i,j}$, respectively. Since Y is nearly uniform-distributed in each cell, the j-th diagonal elements in (19) is approximated by:

$$\frac{1}{\Delta} \int_{\hat{y}_{i,j}-\frac{\Delta}{2}}^{\hat{y}_{i,j}+\frac{\Delta}{2}} (y_j - \hat{y}_{i,j})^2 dy_j = \frac{\Delta^2}{12}. \tag{20}$$

The element in the j-th row and k-th column ($j \neq k$) in (19) is approximated by

$$\left(\frac{1}{\Delta} \int_{\hat{y}_{i,j}-\frac{\Delta}{2}}^{\hat{y}_{i,j}+\frac{\Delta}{2}} (y_j - \hat{y}_{i,j}) dy_j\right) \left(\frac{1}{\Delta} \int_{\hat{y}_{i,k}-\frac{\Delta}{2}}^{\hat{y}_{i,k}+\frac{\Delta}{2}} (y_k - \hat{y}_{i,k}) dy_k\right) = 0. \tag{21}$$

Thus,

$$\mathbb{E}_Y\{(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y} = \hat{y}_i\} \approx \frac{\Delta^2}{12} I_K. \tag{22}$$

Denote $A = \mathbb{E}_Y\{(Y - \hat{Y})(Y - \hat{Y})^T \mid \hat{Y}\}$, $A \approx \frac{\Delta^2}{12} I_K$. Furthermore, we have

$$A_{jk} \approx \begin{cases} \frac{\Delta^2}{12}, & j = k \\ 0, & j \neq k \end{cases} \tag{23}$$

Substitute in (17),

$$\begin{aligned} &\mathbb{E}_Y\{(Y - \hat{Y})^T W(\hat{Y})(Y - \hat{Y}) \mid \hat{Y}\} \\ &= \mathrm{tr}[W(\hat{Y})A] \\ &= \sum_{j=1}^{K} W_{jj}(\hat{Y}) A_{jj} + \sum_{j \neq k} W_{jk}(\hat{Y}) A_{kj} \\ &\approx \frac{\Delta^2}{12} \sum_{j=1}^{K} W_{jj}(\hat{Y}) \\ &= \frac{\Delta^2}{12} \mathrm{tr}[W(\hat{Y})], \end{aligned} \tag{24}$$

where $W_{jk}(\hat{Y})$ denotes the element in the j-th row and k-th column in $W(\hat{Y})$.

So

$$D_{quant} \approx \frac{\Delta^2}{12}\mathbb{E}_{\hat{Y}}\{\mathrm{tr}[W(\hat{Y})]\}. \tag{25}$$

$$\mathbb{E}_{\hat{Y}}\{\mathrm{tr}[W(\hat{Y})]\} = \sum_i P(Y \in \mathcal{C}_i)\,\mathrm{tr}[W(\hat{y}_i)]. \tag{26}$$

Denote $\bar{f}_Y(y) = P(Y \in \mathcal{C}_i)/\Delta^K, \forall y \in \mathcal{C}_i$. Under high-resolution condition and the assumption that the first and second-order derivatives of $g(\cdot)$ are bounded, $\bar{f}_Y(y) \approx f_Y(y)$ and $\mathrm{tr}[W(\hat{y}_i)] \approx \mathrm{tr}[W(y)], \forall y \in \mathcal{C}_i$. Hence

$$\begin{aligned}\mathbb{E}_{\hat{Y}}\{\mathrm{tr}[W(\hat{Y})]\} &= \sum_i \int_{\mathcal{C}_i} \bar{f}_Y(y)\,\mathrm{tr}[W(\hat{y}_i)]dy \\ &\approx \int f_Y(y)\,\mathrm{tr}[W(y)]dy.\end{aligned} \tag{27}$$

$$\begin{aligned}D_{quant} &\approx \frac{\Delta^2}{12}\int f_Y(y)\,\mathrm{tr}[W(y)]dy \\ &= \frac{\Delta^2}{12}\mathbb{E}\{\mathrm{tr}[W(Y)]\}.\end{aligned} \tag{28}$$

Under high-resolution condition, the rate of the quantizer is

$$R(\Phi, Q_1) = h(Y) - K\log\Delta. \tag{29}$$

So the distortion-rate performance of the RPQ scheme with estimator g is

$$D \approx D_0(\Phi) + \frac{\mathbb{E}\{\mathrm{tr}[W(Y)]\}}{12}2^{\frac{2}{K}h(Y)}2^{-\frac{2}{K}R}. \tag{30}$$

For independent marginal entropy coding,

$$D \approx D_0(\Phi) + \frac{\mathbb{E}\{\mathrm{tr}[W(Y)]\}}{12}2^{\frac{2}{K}\sum_{i=1}^K h(Y_i)}2^{-\frac{2}{K}R}. \tag{31}$$

Appendix B

Since S and Y are jointly Gaussian, the relationship between the two random variables can be written as $S = BY + N$. N is a Gaussian noise independent of Y and follows $\mathcal{N}(0, \Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T)$, where Σ_S, Σ_Y and Σ_{SY} represents the covariance matrices of S, Y and their cross-covariance matrix, respectively. So we have

$$D_0(\Phi) = \mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T]. \tag{32}$$

Specifically in the source described in Section IV, we have $\Sigma_S = 1$, $\Sigma_{SY} = \Sigma_{SX}\Phi^T$ and $\Sigma_Y = \Phi\Sigma_X\Phi^T$. Evidently, we have $\Sigma_{SX} = h^T$ and $\Sigma_X = hh^T + \sigma_W^2 I_n$ in this source. Therefore, $\Sigma_{SY} = h^T\Phi^T$. So we have

$$D_0(\Phi) = 1 - h^T\Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi h. \tag{33}$$

Denote $M = \Phi^T(\Phi\Sigma_X\Phi^T)^{-1}\Phi$. For Σ_Y, we have

$$\Sigma_Y = \Phi hh^T\Phi^T + \sigma_W^2\Phi\Phi^T. \tag{34}$$

Denote $\phi = \Phi h$. So

$$\Sigma_Y = \phi\phi^T + \sigma_W^2\Phi\Phi^T. \tag{35}$$

From Sherman-Morrison formula, we have

$$\begin{aligned}\Sigma_Y^{-1} &= (\sigma_W^2\Phi\Phi^T + \Phi hh^T\Phi^T)^{-1} \\ &= \frac{1}{\sigma_W^2}A^{-1} - \frac{A^{-1}\phi\phi^T A^{-1}}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi},\end{aligned} \tag{36}$$

where $A = \Phi\Phi^T$. So

$$M = \Phi^T(\frac{1}{\sigma_W^2}A^{-1} - \frac{A^{-1}\phi\phi^T A^{-1}}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi})\Phi. \tag{37}$$

Further, we have

$$\begin{aligned}h^T M h &= \frac{1}{\sigma_W^2}\phi^T A^{-1}\phi - \frac{\phi^T A^{-1}\phi\phi^T A^{-1}\phi}{\sigma_W^4 + \sigma_W^2\phi^T A^{-1}\phi} \\ &= \frac{\phi^T A^{-1}\phi}{\sigma_W^2 + \phi^T A^{-1}\phi}.\end{aligned} \tag{38}$$

Denote $t = \phi^T A^{-1}\phi$, we have

$$\begin{aligned}\mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T] &= 1 - h^T M h \\ &= 1 - \frac{t}{\sigma_W^2 + t} \\ &= \frac{\sigma_W^2}{\sigma_W^2 + t}.\end{aligned} \tag{39}$$

Denote $G = \Phi^T A^{-1}\Phi$, we have $t = h^T G h$. Since G is symmetric and idempotent, it follows that

$$\begin{aligned}t &= h^T G^T G h \\ &= \|Gh\|_2^2.\end{aligned} \tag{40}$$

For an arbitrary non-zero vector v, consider $\frac{\|Gv\|_2^2}{\|v\|_2^2}$. Without loss of generality, assume that $\|v\|_2^2 = 1$. Denote $e_1 = (1, 0, \ldots, 0)^T$. Then v can be expressed as $v = Ue_1$, where U is an orthogonal matrix. So we have

$$\begin{aligned}\|Gv\|_2^2 &= \|GUe_1\|_2^2 \\ &= e_1^T U^T G U e_1 \\ &= e_1^T(\Phi U)^T((\Phi U)(\Phi U)^T)^{-1}(\Phi U)e_1.\end{aligned} \tag{41}$$

Denote the row vectors of Φ as $\Phi_1^T, \Phi_2^T, \ldots, \Phi_K^T$, i.e.,

$$\Phi = \begin{pmatrix}\Phi_1^T \\ \Phi_2^T \\ \vdots \\ \Phi_K^T\end{pmatrix}. \tag{42}$$

So

$$\Phi U = \begin{pmatrix}\Phi_1^T U \\ \Phi_2^T U \\ \vdots \\ \Phi_K^T U\end{pmatrix}. \tag{43}$$

For the i-th row vector of Φ, $\Phi_i \sim \mathcal{N}(\mathbf{0}, \frac{1}{K}I_n)$. So

$$\begin{aligned}\mathbb{E}\{U^T\Phi_i\Phi^T U\} &= \frac{1}{K}U^T U \\ &= \frac{1}{K}I_n.\end{aligned} \tag{44}$$

Hence ΦU has the same distribution as Φ. Combined with (41), it follows that $\|Ge_1\|_2^2$ and $\|Gv\|_2^2$ are identically distributed. Therefore, $\frac{t}{\|h\|_2^2} = \frac{\|Gh\|_2^2}{\|h\|_2^2}$ and $\frac{\|Ge_1\|_2^2}{\|e_1\|_2^2} = \|Ge_1\|_2^2$ are identically distributed. So for simplicity and without loss of generality, we study the distribution of $\|Ge_1\|_2^2$ in the following derivation.

$$\begin{aligned}\|Ge_1\|_2^2 &= e_1^T G^T G e_1 \\ &= e_1 \Phi^T(\Phi\Phi^T)^{-1}\Phi e_1.\end{aligned} \tag{45}$$

Define a random matrix $\Psi = \sqrt{K}\Phi$. Then Ψ has i.i.d. entries following $\mathcal{N}(0,1)$, and $\Psi^T(\Psi\Psi^T)^{-1}\Psi = \Phi^T(\Phi\Phi^T)^{-1}\Phi = G$. So

$$e_1^T G e_1 = e_1^T \Psi^T(\Psi\Psi^T)^{-1}\Psi e_1. \tag{46}$$

Define $a = \Psi e_1$, which is simply the first column of Ψ. Hence

$$e_1^T G e_1 = a^T(\Psi\Psi^T)^{-1} a. \tag{47}$$

Partition Ψ as

$$\Psi = \left(\begin{array}{ccc} a & \vdots & W \end{array}\right), \tag{48}$$

where the size of W is $K \times (n-1)$. $\Psi\Psi^T = aa^T + WW^T$. Denote $V = WW^T$. From Sherman-Morrison formula, we have

$$(aa^T + V)^{-1} = V^{-1} - \frac{V^{-1}aa^TV^{-1}}{1 + a^TV^{-1}a}. \tag{49}$$

$$\begin{aligned} a^T(aa^T+V)^{-1}a &= a^TV^{-1}a - \frac{(a^TV^{-1}a)^2}{1+a^TV^{-1}a} \\ &= \frac{a^TV^{-1}a}{1+a^TV^{-1}a}. \end{aligned} \tag{50}$$

$V \sim W_K(I_K, n-1)$ (W_K denotes the Wishart distribution) and $a \sim \mathcal{N}(\mathbf{0}, I_K)$. So $(n-1)a^TV^{-1}a \sim T^2(K, n-1)$ and $\frac{n-K}{K}a^TV^{-1}a \sim F(K, n-K)$. Since $\frac{n-K}{K}a^TV^{-1}a \sim F(K, n-K)$ follows an $F(K, n-K)$ distribution, it can be expressed as the ratio of two independent chi-square random variables normalized by their respective degrees of freedom, i.e.,

$$\frac{n-K}{K}a^TV^{-1}a = \frac{\frac{U_1}{K}}{\frac{U_2}{n-K}} = \frac{n-K}{K}\frac{U_1}{U_2}, \tag{51}$$

where $U_1 \sim \chi^2(K)$, $U_2 \sim \chi^2(n-K)$, and U_1 and U_2 are independent. After simplification, we obtain:

$$a^TV^{-1}a = \frac{U_1}{U_2}. \tag{52}$$

Hence

$$\|Ge_1\|_2^2 = \frac{U_1}{U_1+U_2}. \tag{53}$$

Since $t = \|Gh\|_2^2$ has the same distribution as $\|h\|_2^2\|Ge_1\|_2^2$, letting $\|Ge_1\|_2^2 = l$, with probability density function $f_L(\cdot)$. Denote $s = t/\|h\|_2^2$. Then s has the same distribution as l. Denote the distribution of t and s as $f_T(\cdot)$ and $f_S(\cdot)$, respectively. Thus $f_S(\cdot) = f_L(\cdot)$ and $f_T(t) = \frac{1}{\|h\|_2^2}f_S(s)$. So we have

$$\begin{aligned} &\mathbb{E}\{\mathrm{tr}[\Sigma_S - \Sigma_{SY}\Sigma_Y^{-1}\Sigma_{SY}^T]\} \\ &= \mathbb{E}\{\frac{\sigma_W^2}{\sigma_W^2 + t}\} \\ &= \int_0^{\|h\|_2^2} f_T(t)\frac{\sigma_W^2}{t+\sigma_W^2}dt \\ &= \int_0^{\|h\|_2^2} \frac{1}{\|h\|_2^2} f_S(s)\frac{\sigma_W^2}{\|h\|_2^2 s+\sigma_W^2}d(\|h\|_2^2 s) \\ &= \int_0^1 f_S(s)\frac{\sigma_W^2}{\|h\|_2^2 s+\sigma_W^2}ds \\ &= \int_0^1 f_L(l)\frac{\sigma_W^2}{\|h\|_2^2 l+\sigma_W^2}dl. \end{aligned} \tag{54}$$

Denote the joint probability density function of U_1 and U_2 by $f_{U_1,U_2}(\cdot)$. We have

$$f_{U_1,U_2}(u_1,u_2) = \frac{1}{2^{\frac{K}{2}}\Gamma(\frac{K}{2})}u_1^{\frac{K}{2}-1}e^{-\frac{u_1}{2}}\frac{1}{2^{\frac{n-K}{2}}\Gamma(\frac{n-K}{2})}u_2^{\frac{n-K}{2}-1}e^{-\frac{u_2}{2}} \tag{55}$$

We have the following transformation:

$$l = \frac{u_1}{u_1+u_2} \tag{56}$$

and

$$w = u_1 + u_2. \tag{57}$$

The inverse transformation is:

$$u_1 = lw \tag{58}$$

and

$$u_2 = (1-l)w. \tag{59}$$

The Jacobian determinant is

$$J = \begin{vmatrix} \frac{\partial u_1}{\partial l} & \frac{\partial u_1}{\partial w} \\ \frac{\partial u_2}{\partial l} & \frac{\partial u_2}{\partial w} \end{vmatrix} = \begin{vmatrix} w & l \\ -w & 1-l \end{vmatrix} = w. \tag{60}$$

Hence

$$f_{L,W}(l,w) = f_{U_1,U_2}(u_1,u_2)\,|J| = f_{U_1,U_2}(lw,(1-l)w)w. \tag{61}$$

Combined with (55), we have

$$f_{L,W}(l,w) = \frac{1}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}}l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}w^{\frac{n}{2}-1}e^{-\frac{w}{2}}. \tag{62}$$

The marginal pdf of l is obtained by integrating the joint pdf with respect to w:

$$\begin{aligned} f_L(l) &= \int_0^{+\infty} f_{L,W}(l,w)dw \\ &= \frac{l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}}\int_0^{+\infty} w^{\frac{n}{2}-1}e^{-\frac{w}{2}}dw \\ &= \frac{l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})2^{\frac{n}{2}}}2^{\frac{n}{2}}\Gamma(\frac{n}{2}) \\ &= \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{K}{2})\Gamma(\frac{n-K}{2})}l^{\frac{K}{2}-1}(1-l)^{\frac{n-K}{2}-1}. \end{aligned} \tag{63}$$

Hence

$$\mathbb{E}\{D_0(\Phi) \mid K\} = \mathbb{E}\{\frac{\sigma_W^2}{\sigma_W^2 + \|h\|_2^2 l}\}, l \sim \mathrm{Beta}(\frac{K}{2}, \frac{n-K}{2}). \tag{64}$$

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